

Digestive System

1. Overview
2. Anatomy & Functions
 - GI tract
 - Accessory structures
3. Mechanisms – digestion & absorption
4. Regulation – 3 hormones, enteric nervous system

Terms

1. **Cell respiration:** biochemical pathway for ATP production (*glycolysis + citric acid cycle + electron transport*)
2. **Metabolism:** sum of all chemical reactions in the body cells (*catabolism + anabolism of lipid, protein & carbohydrate*)
3. **Digestion:** occurs in a tube through the middle of the body, extracellular
 - **Digest:** convert **polymers** → **monomers/subunits**
 - **Protein** → **amino acid**
 - **Starch** → **glucose**
 - **Triglycerides** → **fatty acids, glycerol**
 - **Absorb:** **monomers/subunits**

Functions

1. Ingest:

- **Nutrients**: chemicals we need and cannot make
- Required for **structure** and **cellular processes** (fuel, synthesis)

2. Digest: polymers to monomers

- Mechanical (*teeth/smooth muscle*)
- Chemical (*enzymes - require specific substrates and conditions*)

3. Move: transport through intestinal tract (*smooth muscle*)

4. Absorb:

- **Blood capillaries**: amino acids, glucose
- **Lymphatic capillaries (lacteals)**: fats

Nutrients

Chemicals we need and cannot make

- 1. Fuel molecules:** glucose and fatty acids (*to make ATP*)
- 2. Amino acids:** to make proteins (*for their many functions*)
- 3. Water:**
 - Solvent
 - Chemical reactions: dehydration synthesis & hydrolysis degradation
 - 70% of body composition is water. Maintains osmolarity, BP
- 4. Minerals:** bone, cofactors, nerve/muscle function, 2nd messengers, oxygen transport
- 5. Vitamins:** coenzymes

Anatomy

gastrointestinal (GI) tract

Oral cavity
digestion

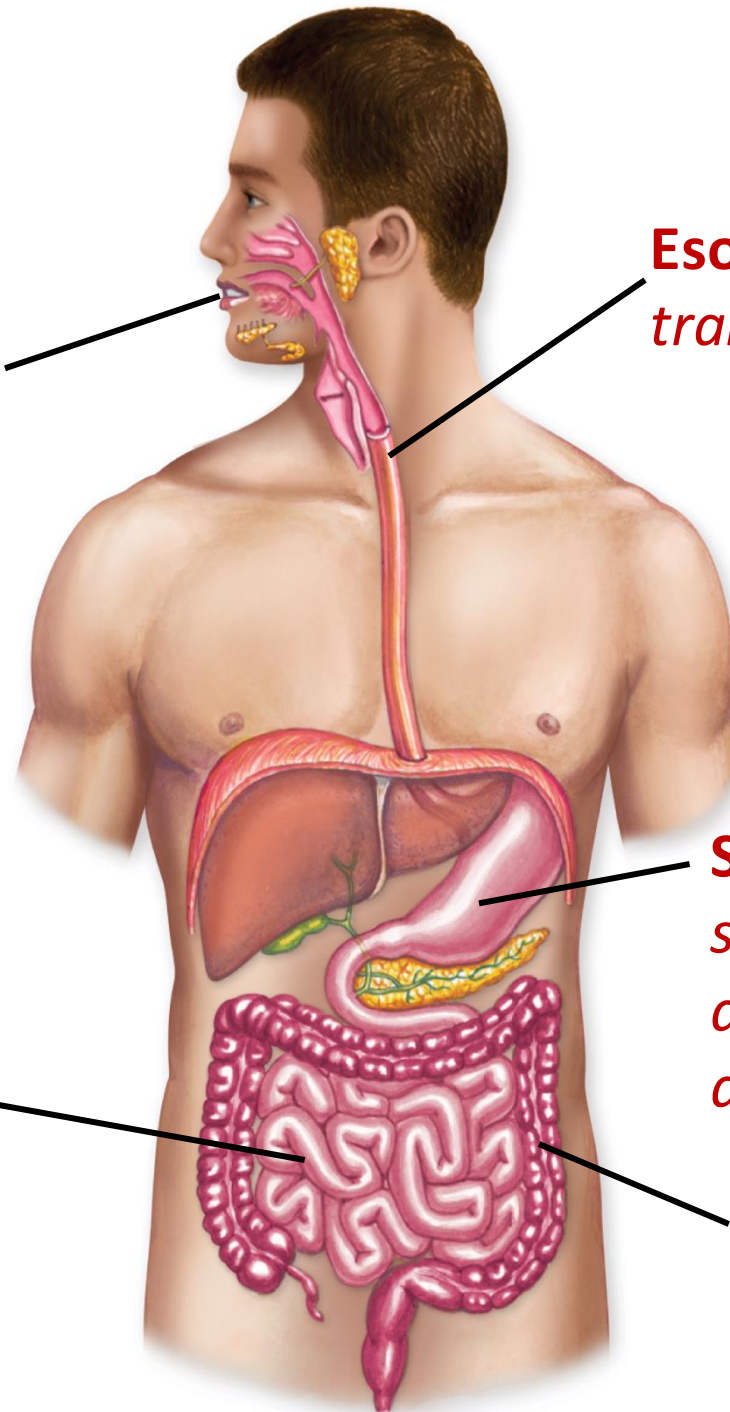
Esophagus
transport tube

Small intestine
digestion & absorption

- Duodenum: 1 ft.
- Jejunum: 5 ft
- Ileum: 7 ft.

Stomach
*storage
digestion
absorption*

Large intestine
*absorption &
solidify feces⁵*



Anatomy

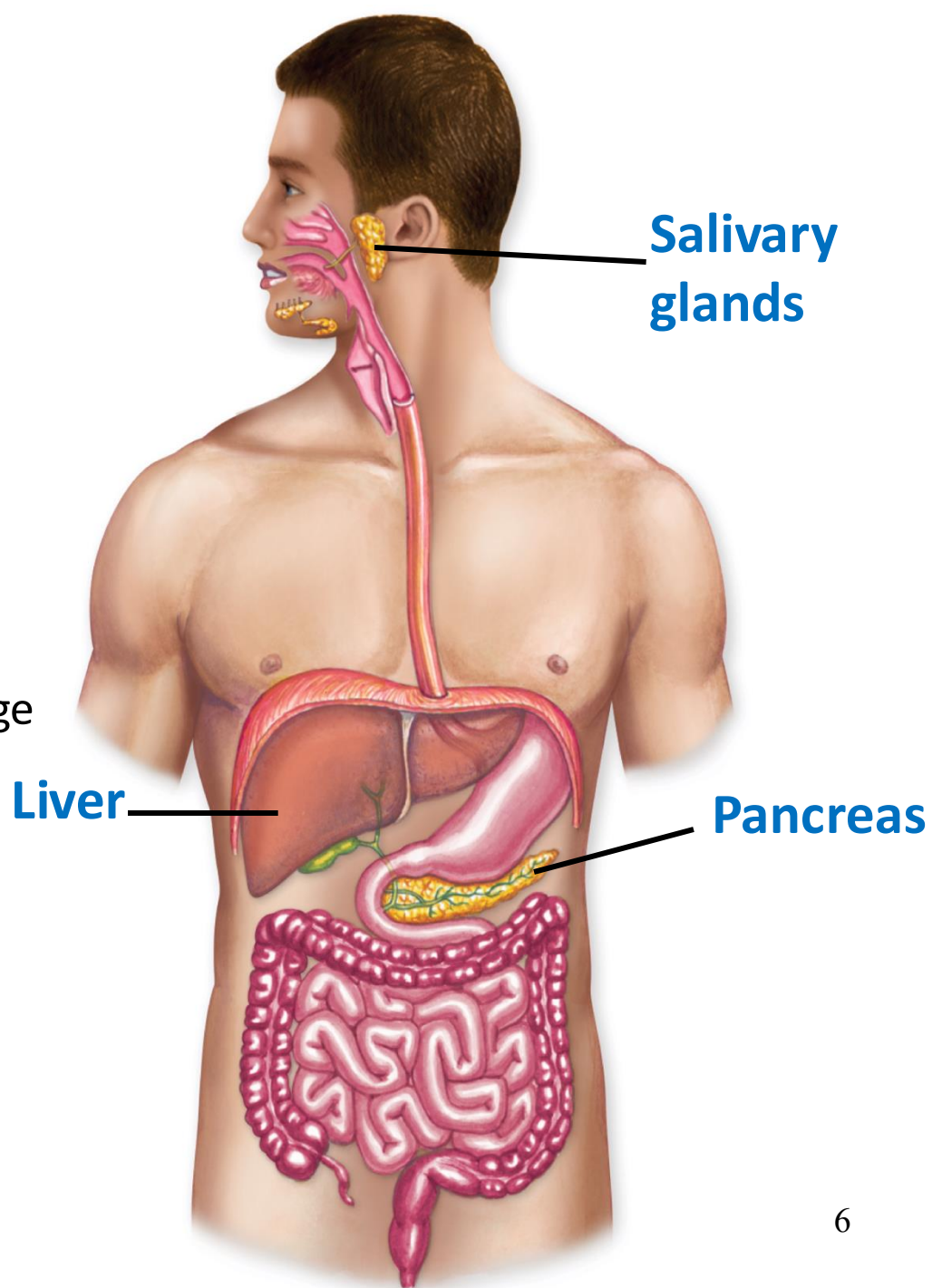
accessory organs

Specialized exocrine organs

- Secrete into digestive tract
(oral cavity & duodenum)

Individual glands

- Single cells and groups of cells
- Secrete into digestive tract
- Lots in colon for smooth passage of solid feces



Generalized Functions

GI tract:

- *Mechanical digestion* (teeth, smooth muscle)
- *Secretion: digestive enzymes and hormones*
- *Enzymatic digestion*
- *Absorption*

Accessory organs:

- *Secrete: into tube*
- *Secrete: digestive juice* (water, ions, digestive enzymes), mucus

Oral cavity

Salivary glands

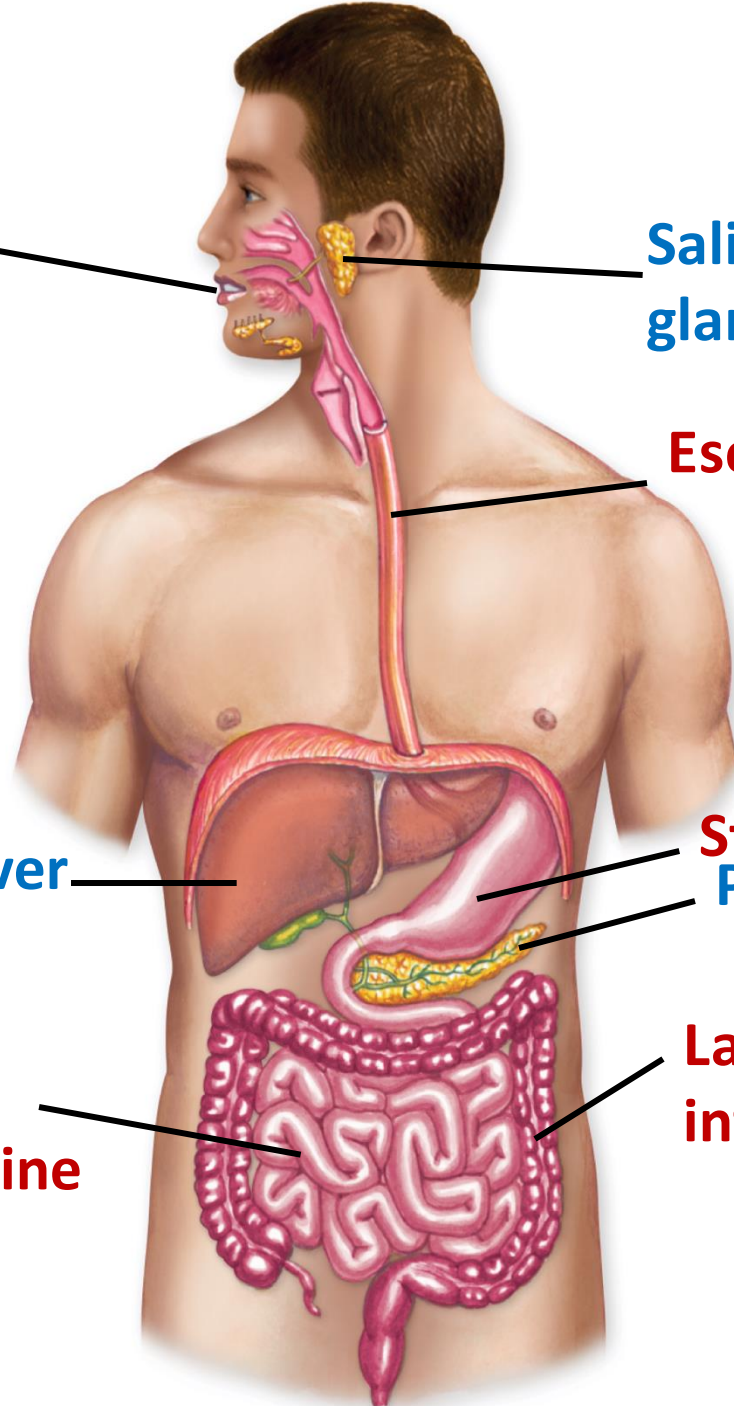
Esophagus

Liver

Stomach
Pancreas

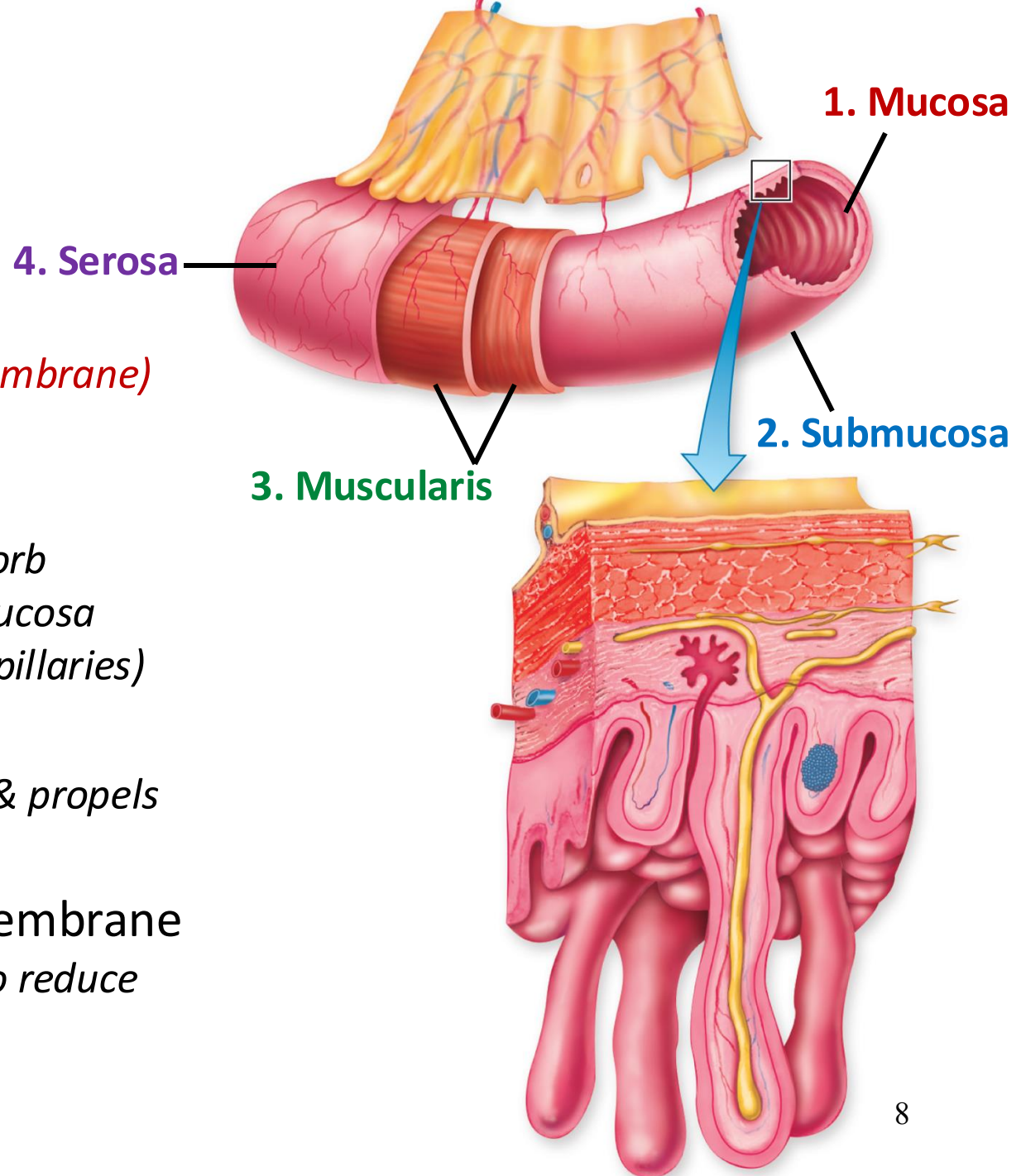
Small intestine

Large intestine



Three layers of GI tract

1. **Mucosa** (*mu*cous membrane)
secretes and absorbs
2. **Submucosa**
*highly vascular to absorb
nutrients that cross mucosa
(lymphatic & blood capillaries)*
3. **Muscularis**
*mechanical digestion & propels
food*
4. **Serosa** = serous membrane
*secretes serous fluid to reduce
friction*



Stomach

anatomy

Function:

- *storage (primary)*
- *mechanically digest*
- *secretes **HCL** & **pepsinogen***
- *secretes **mucus***
- *absorb water & medications*

Folds = rugae

Gastric pits

Mucous cell
mucus

Parietal cell
HCl

Chief cell
pepsinogen

Mucosa

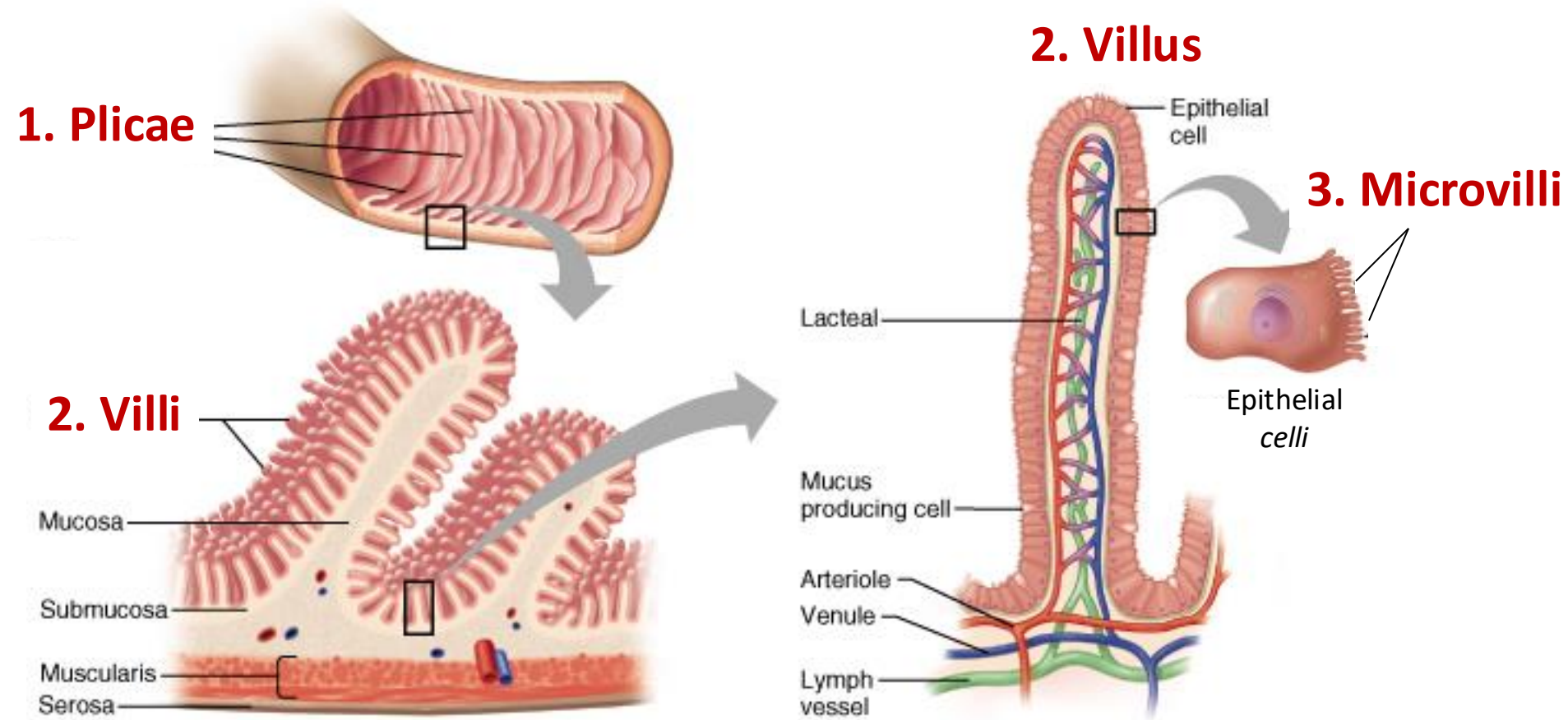
Submucosa

Gastric gland

Small Intestine

anatomy

3 folds → increase SA



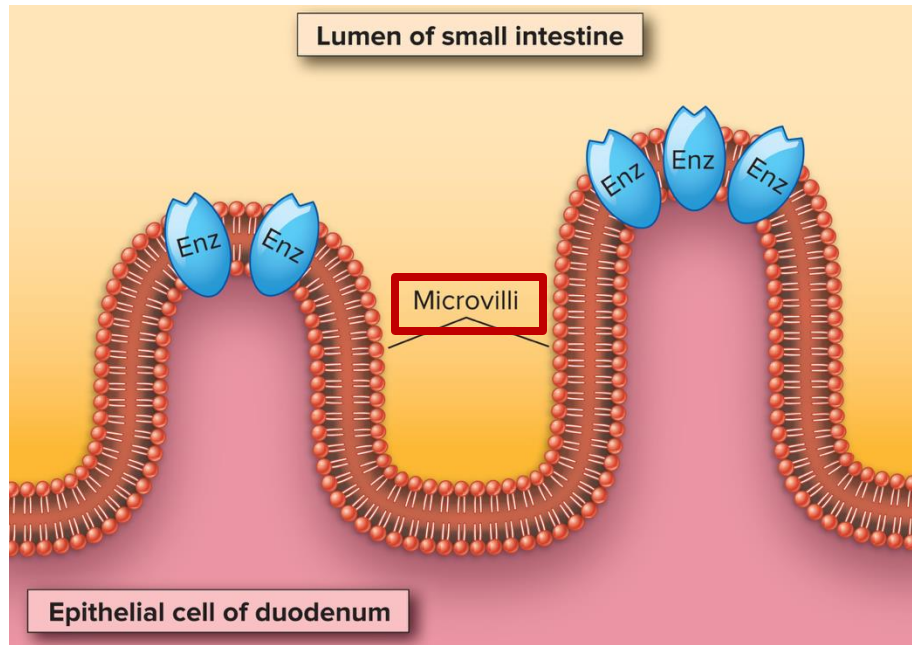
Epithelial cells:

- “Brush border” – fuzzy appearance due to **microvilli**
- Have enzymes anchored to cells → final digestion

Small Intestine

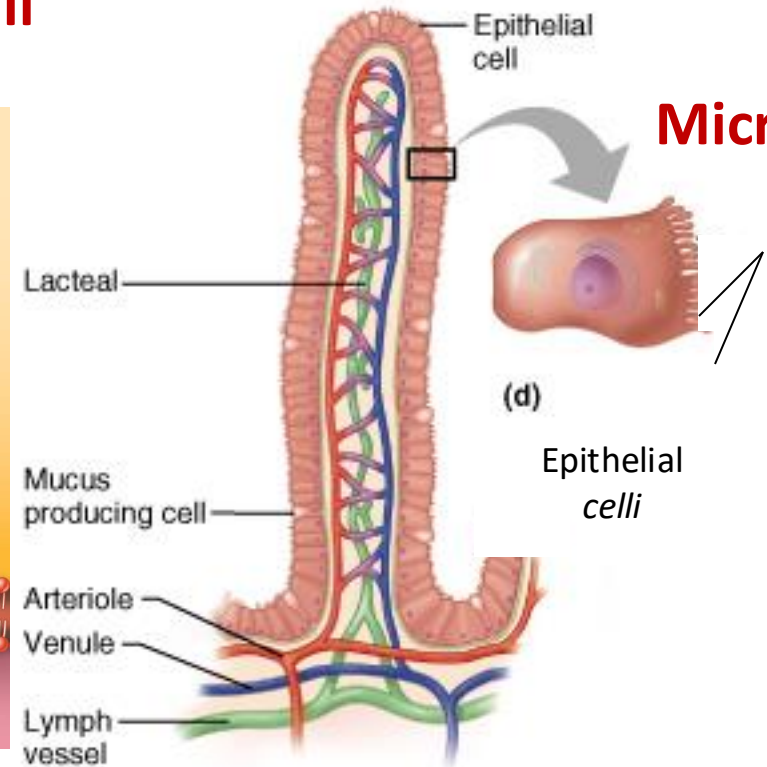
anatomy

Enzymes anchored within **microvilli**



Villus

Microvilli



Functions: primary site for

- **Digestion** of nutrients
- **Absorption** of monomers/subunits; 3 levels of folding increase surface area

Accessory Organ Functions

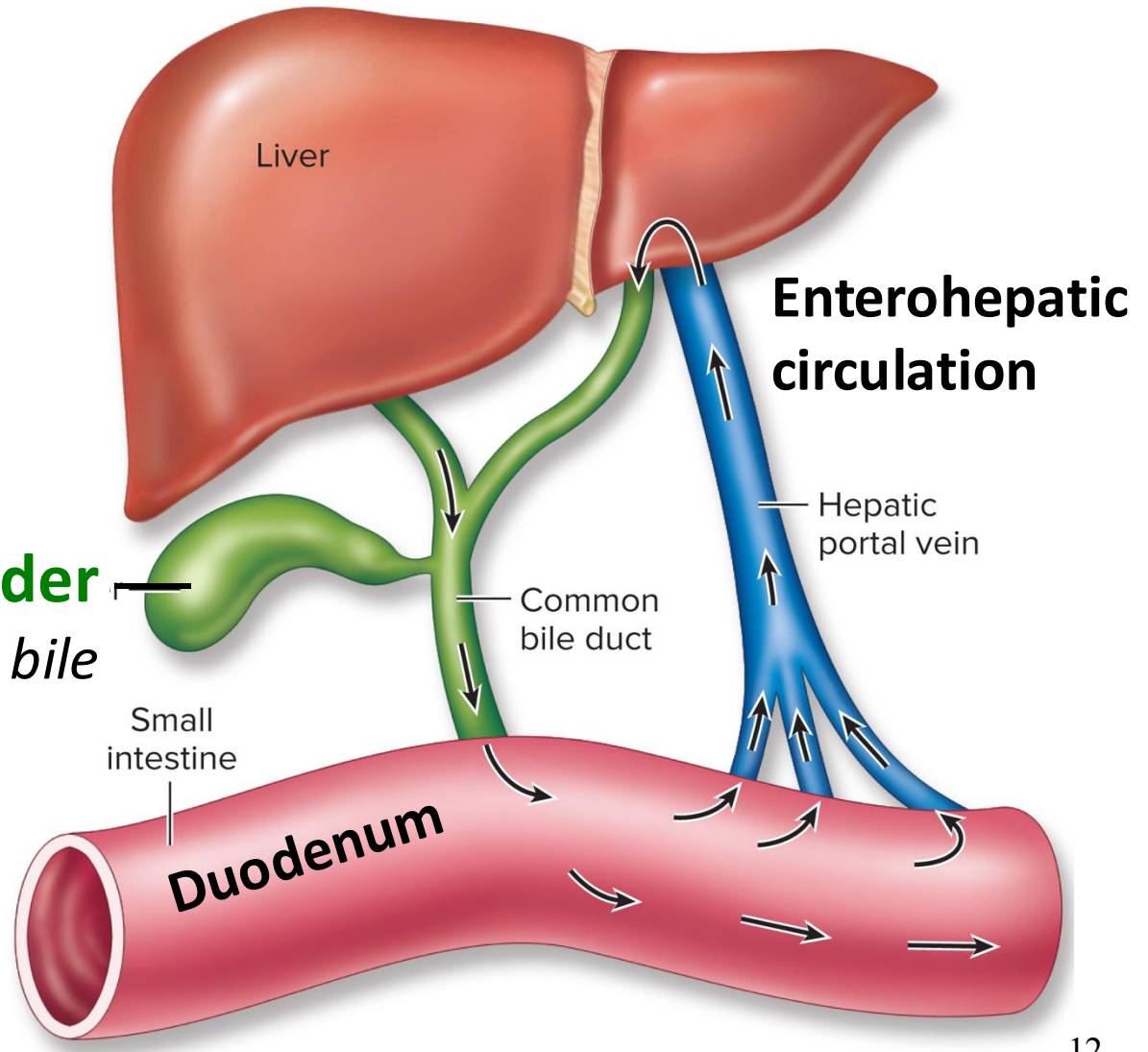
liver & *gall bladder*

Liver:

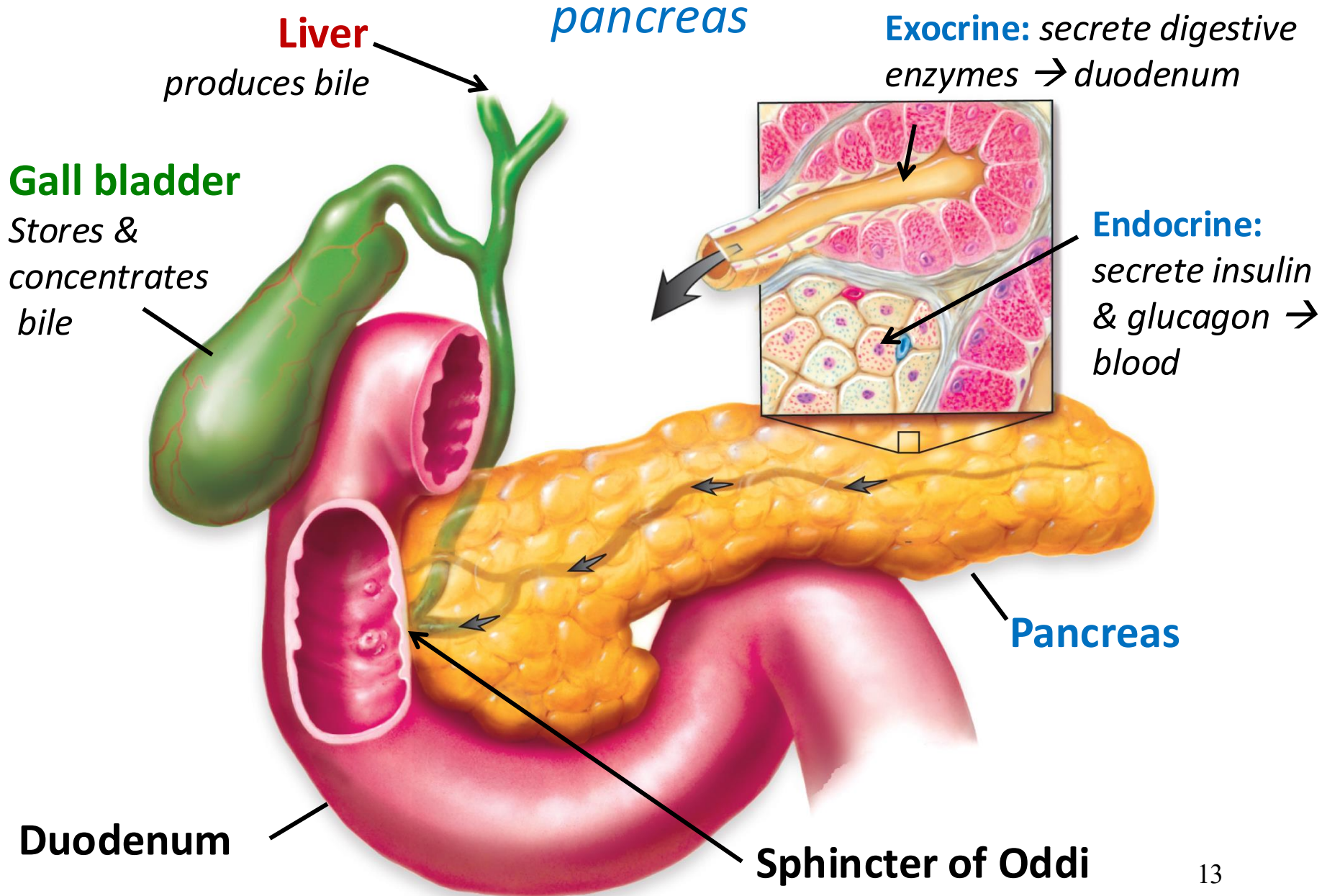
- *produces bile*
 - *emulsify fats*
 - *reabsorbed/reused*

Gall bladder

stores & concentrates bile



Accessory Organ Functions



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Processes of Digestion

1. Motility

peristalsis, segmentation and sphincters

2. Mechanical digestion

teeth & muscle contraction (segmentation)

3. Chemical digestion (enzymes → hydrolysis)

carbohydrates, proteins, fats, (nucleotides)

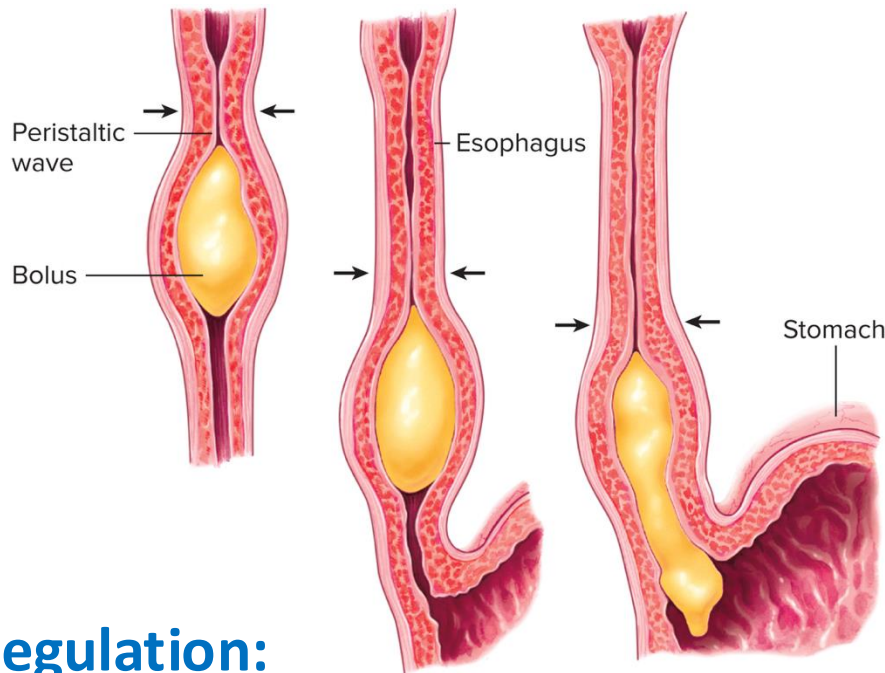
4. Absorption

epithelial transport (aa's & glucose); exocytosis (lipids)

Motility & Its Regulation

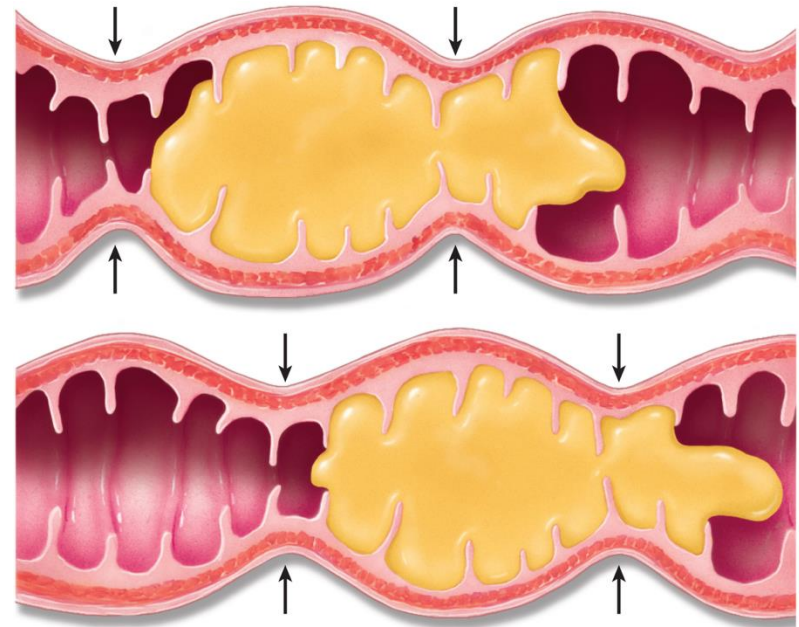
Peristalsis

propels food



Segmentation

mixes & propels food



Regulation:

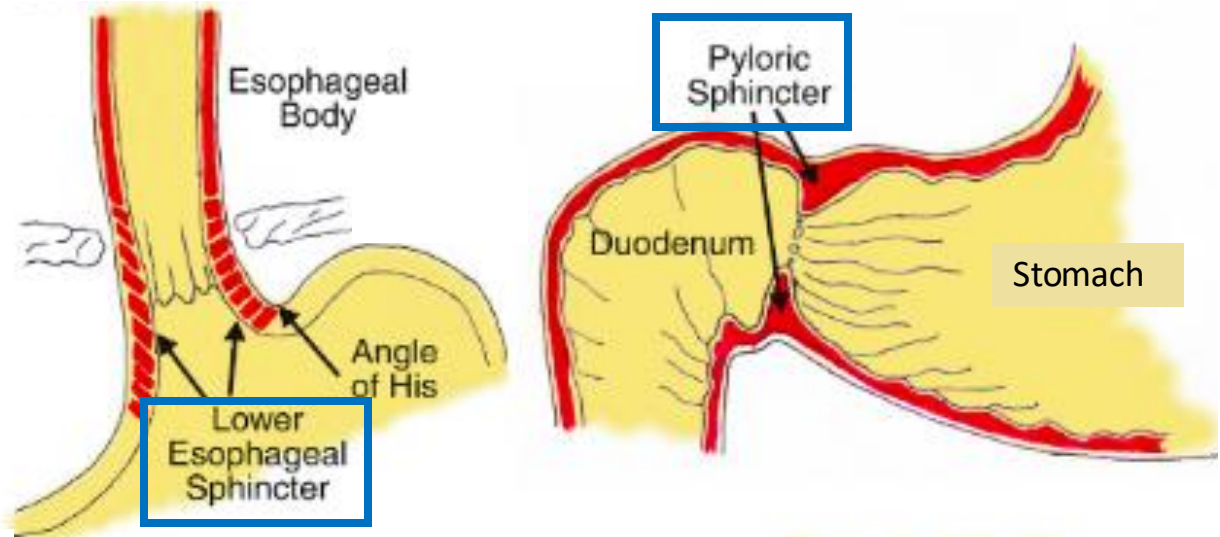
- **Neural control:**
 - **Enteric NS:** local control nervous system independent of brain; 100 million neurons; 2nd brain
 - **Autonomic NS:** sympathetic/parasympathetic NS (50,000 neurons)
- **Hormonal control:** paracrine
- **Pacemaker cell control:** intrinsic rhythm of smooth muscle due to pacemaker activity; gap junctions connect pacemaker cells to smooth muscle.

Sphincters

circular muscle that open/close tubes

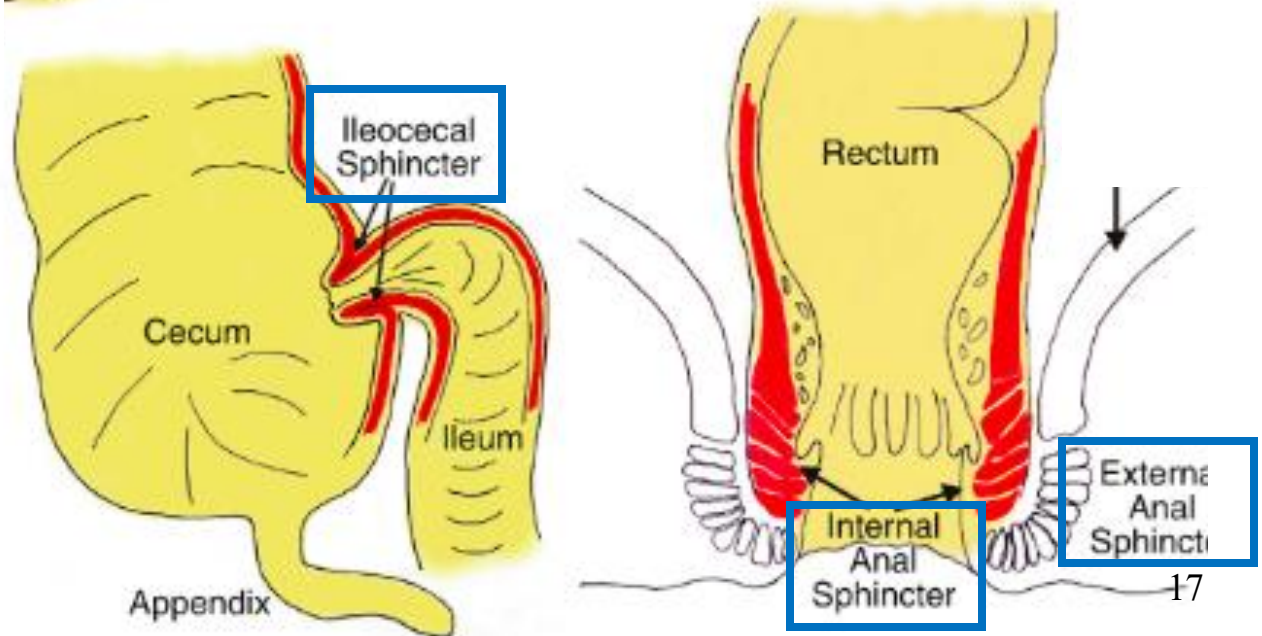
Function:

Regulate passage of **bolus**/**chyme** passage between organs.



Bolus: *food in esophagus*

Chyme: *liquefied bolus; in stomach*



Processes of Digestion

1. Motility

peristalsis, segmentation and sphincters

2. Mechanical digestion

teeth & muscle contraction (segmentation)

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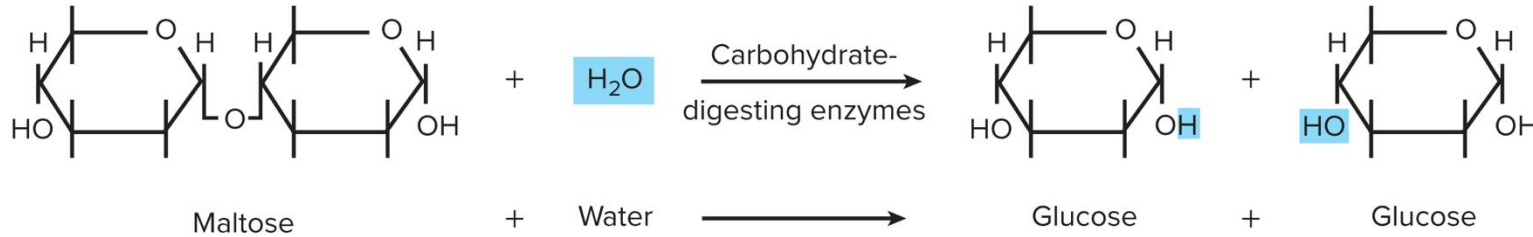
carbohydrates, proteins, fats, (nucleotides)

4. Absorption

epithelial transport (aa's & glucose); exocytosis (lipids)

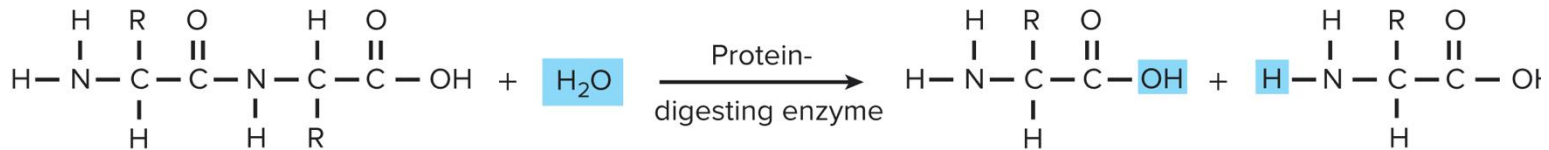
Chemical Digestion is Hydrolysis

Carbohydrate



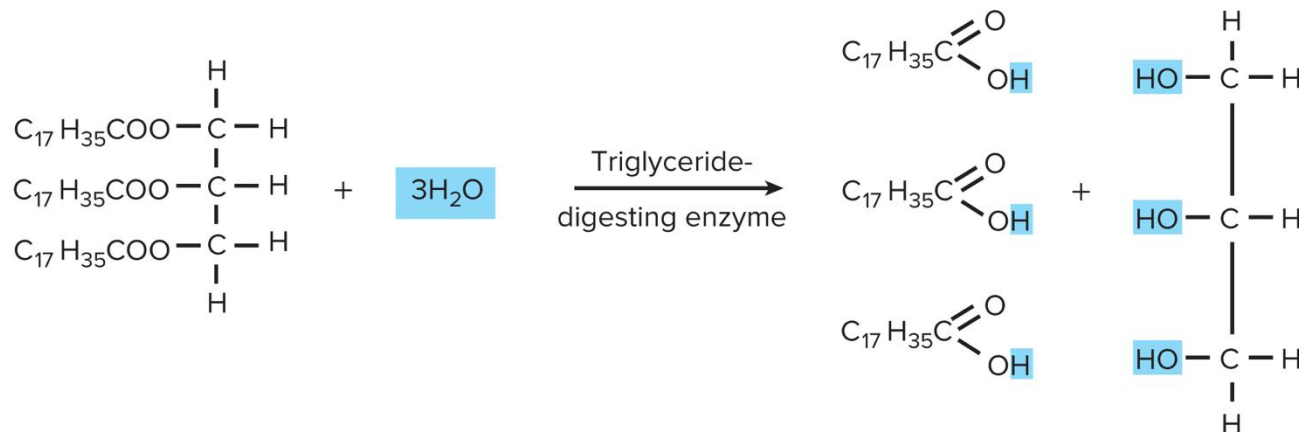
Disaccharide + Water $\longrightarrow$ Monosaccharides

Protein



Peptide (portion of protein molecule) + Water $\longrightarrow$ Amino acid + Amino acid

Lipid



Fat + Water $\longrightarrow$ Fatty acids + Glycerol

Carbohydrate Digestion & Absorption

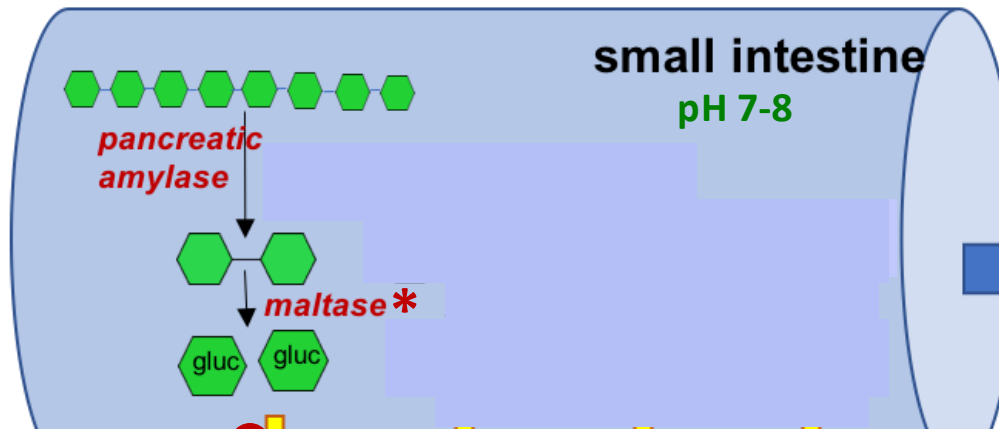
enzymes

Stomach: pH 1-3
no further digestion

Mouth: pH 7-8

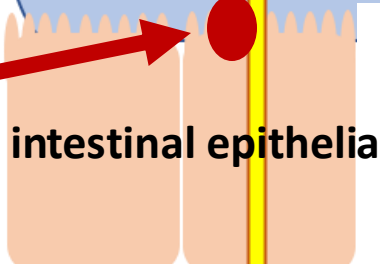
Salivary amylase: *partial digestion to smaller polysaccharide*

Ingest **starch**
(polysaccharide)



Undigested cellulose → to **large intestine** → food for microbiome

*** brush border enzymes**



intestinal epithelia



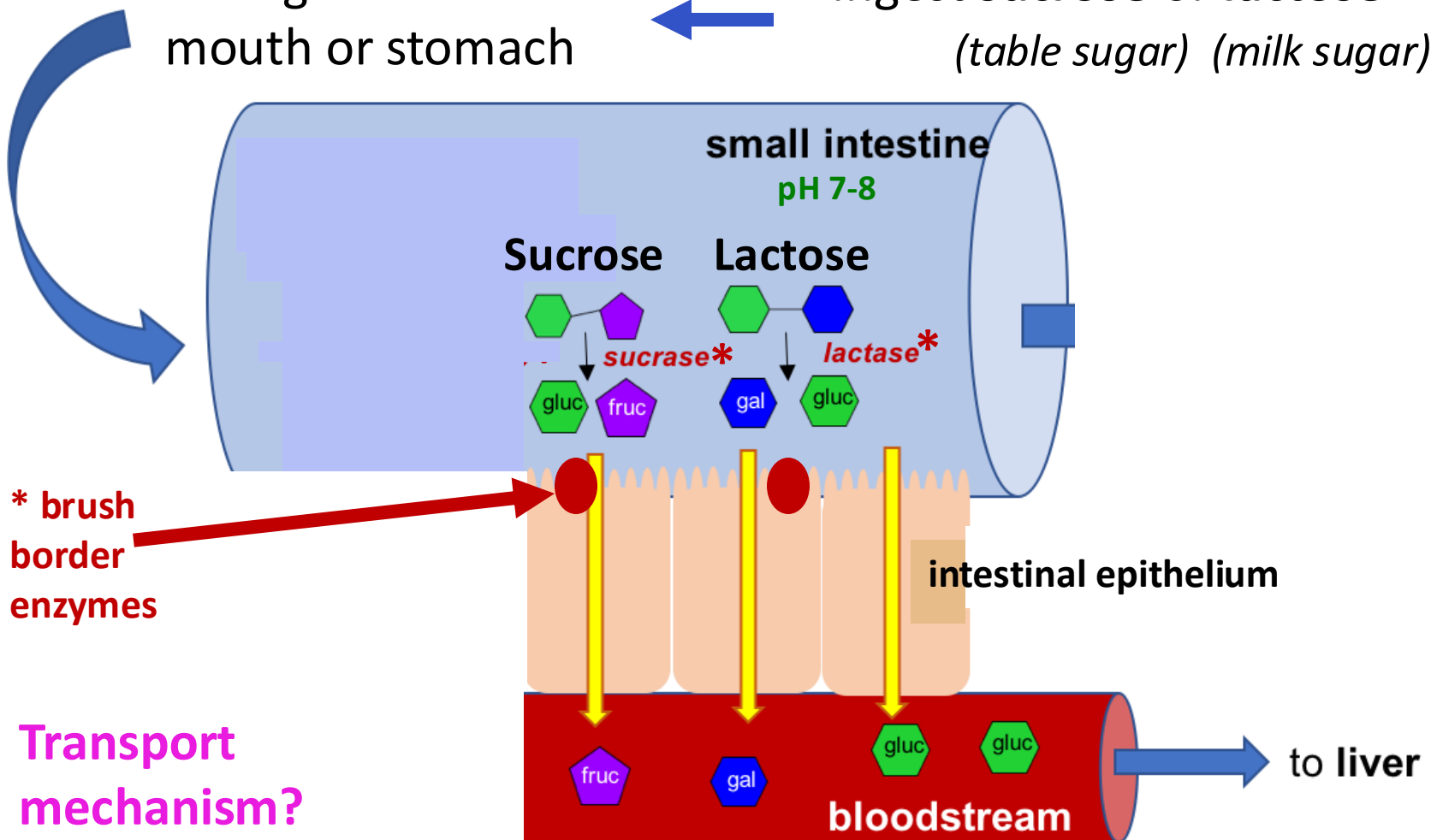
to **liver**

Carbohydrate Digestion & Absorption

enzymes

No digestion in
mouth or stomach

Ingest **sucrose** or **lactose**
(table sugar) (milk sugar)



What is the source of maltose? ...sucrose? ...lactose?

Protein Digestion

enzymes

Mouth
Mechanical digestion

Stomach

$\text{Pepsinogen} + \text{HCl} \rightarrow \text{pepsin}$

Pepsin digests proteins $\rightarrow$ *shorter-chain polypeptides*

Small intestine

- **Pancreatic enzymes:** *trypsin, chymotrypsin, carboxypeptidase*

- **Brush border enzyme:** *aminopeptidase*

Enzymes: digest *short-chain polypeptides* $\rightarrow$ *amino acids, dipeptides & tripeptides*

Intestinal cells (inside)

Dipeptidases & tripeptidases

Digest *dipeptides & tripeptides* $\rightarrow$ *amino acids*

Protein Digestion

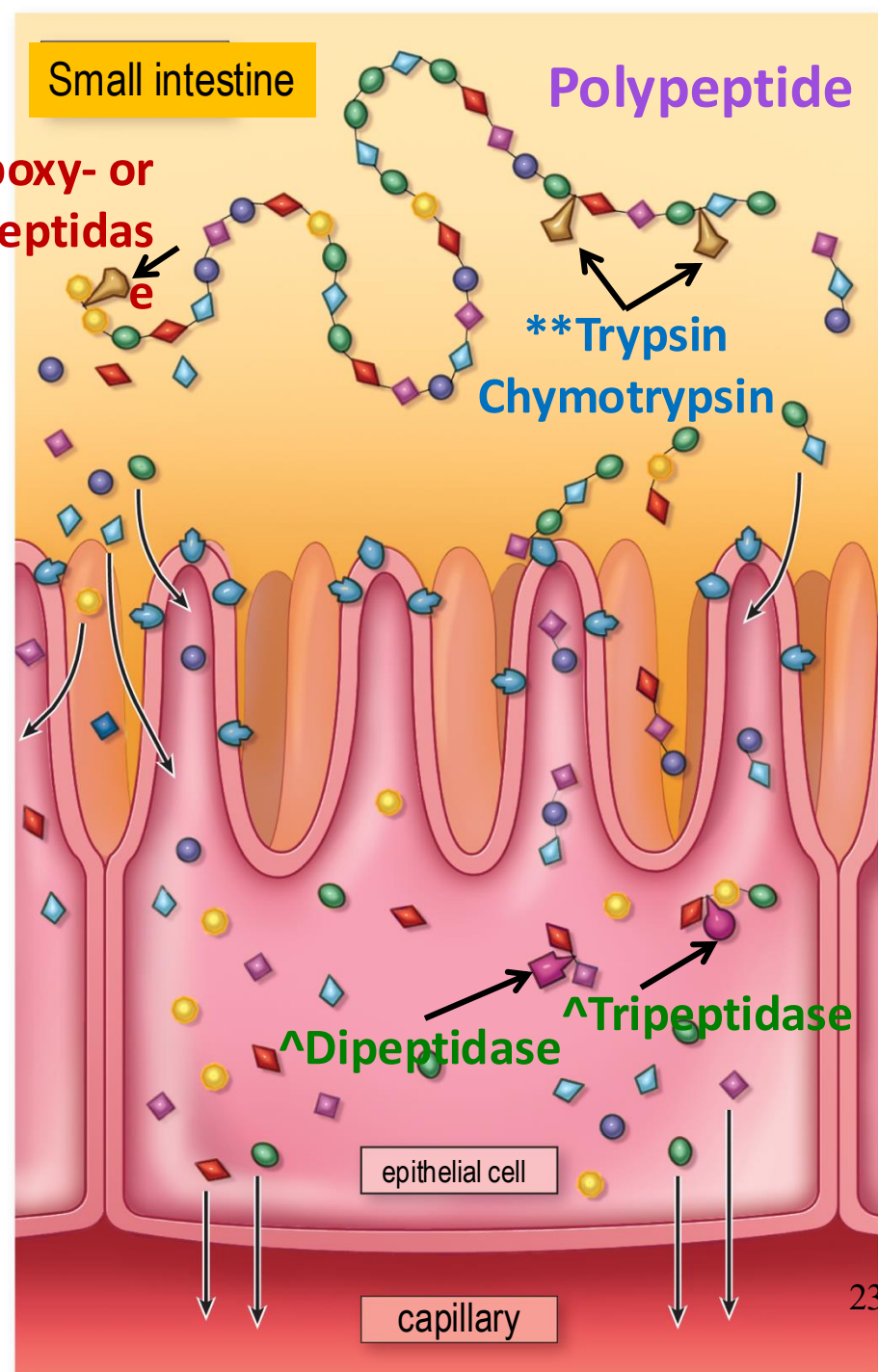
Pepsin (*stomach*) cleaves long proteins into **polypeptides** (*small proteins*) → **small intestine**

*Cleave peptide bonds at **end** of polypeptides

Cleave peptide bonds in **middle of polypeptides

^Cleave di- & tri-peptides to amino acids

Transport mechanism?



Lipid Digestion

lipids are macromolecule (not really polymers)

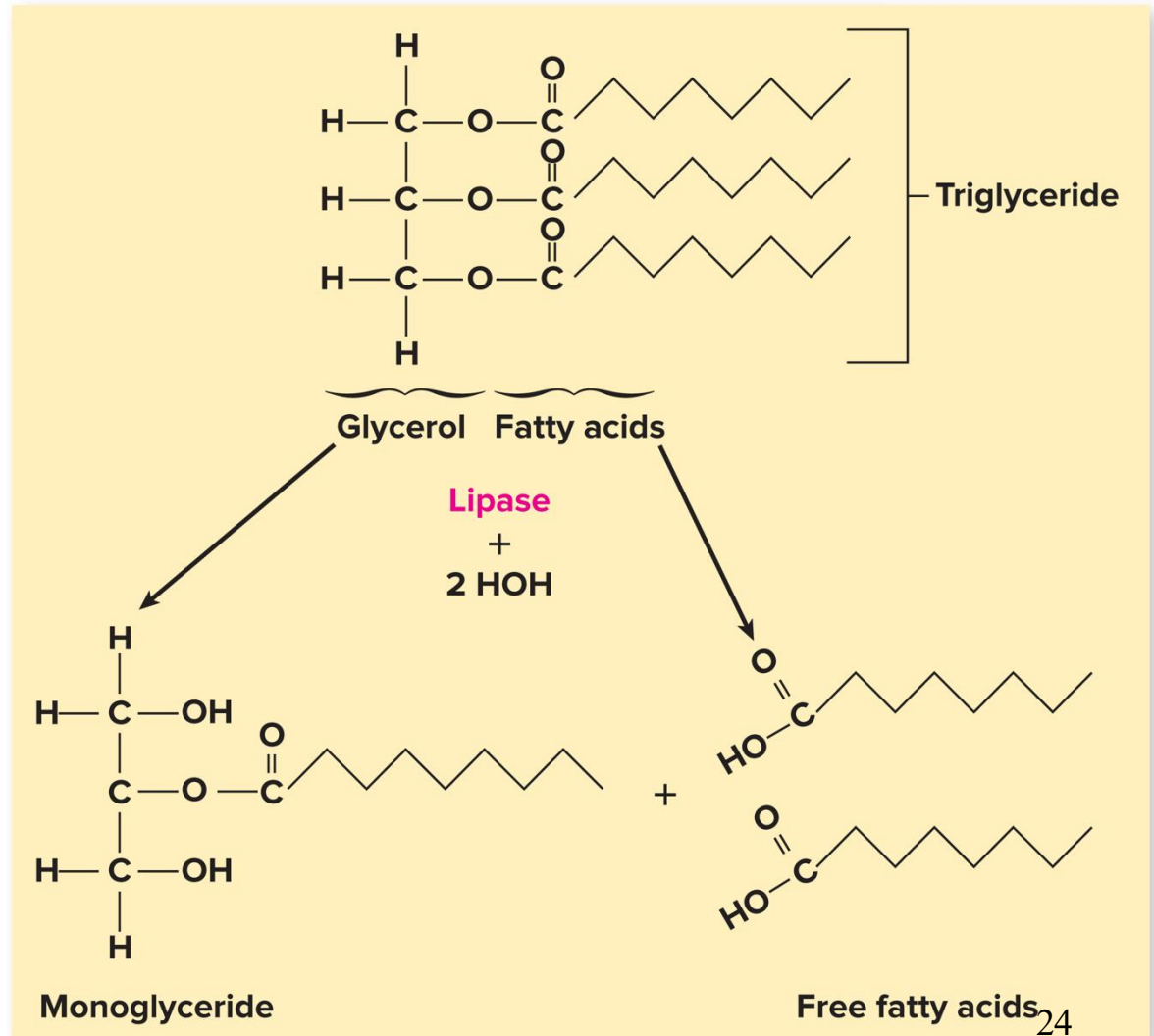
We consume lipids as
triglycerides (TG)

- Fats (*solid*)
- Oils (*liquid*)

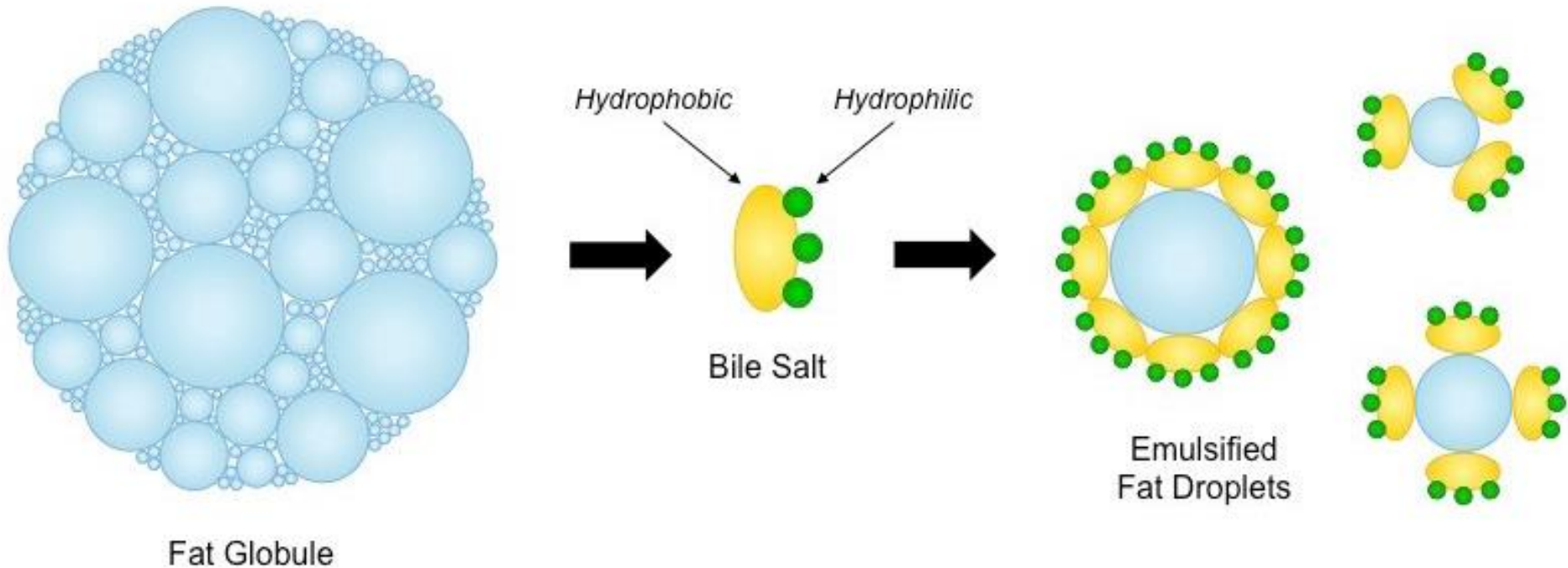
TG's differ in their
fatty acids.

Fatty acids differ in

- Length
- Saturation with H

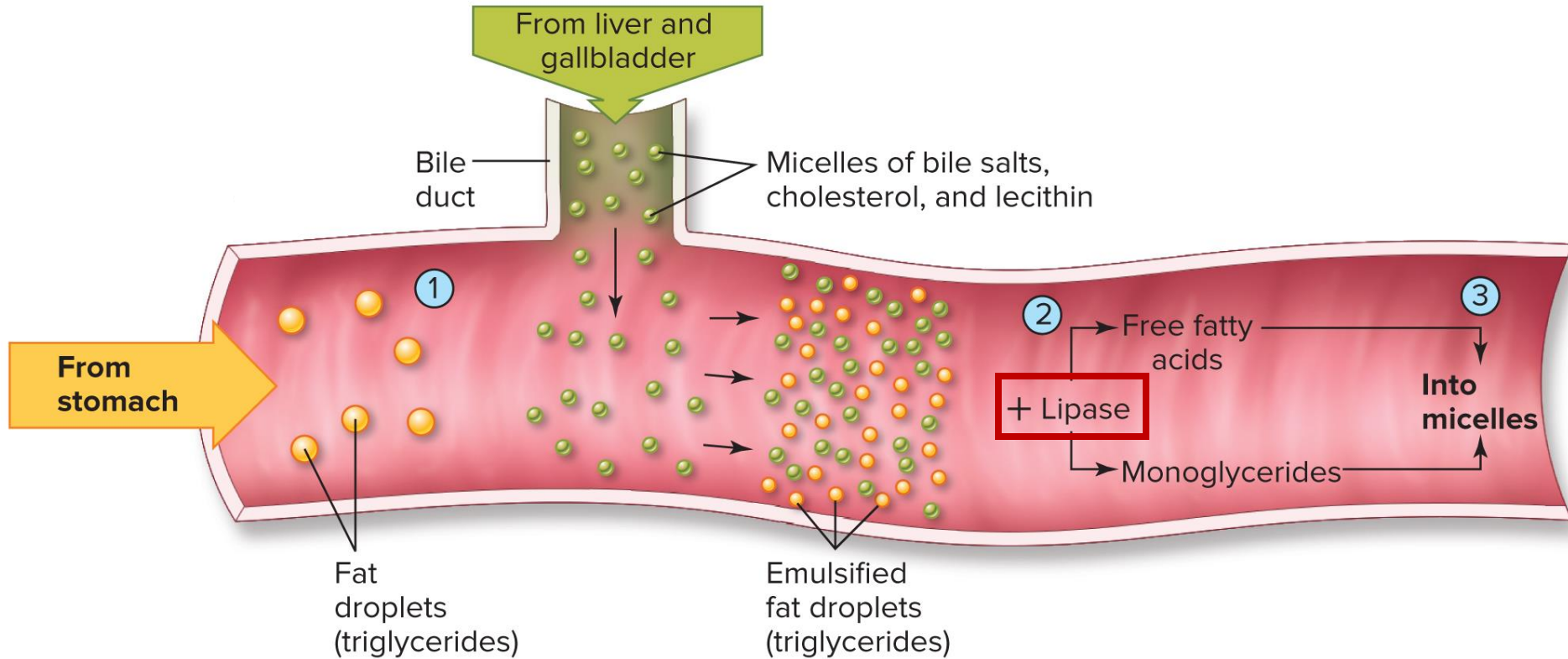


Emulsification by Bile Salts



Lipid Digestion

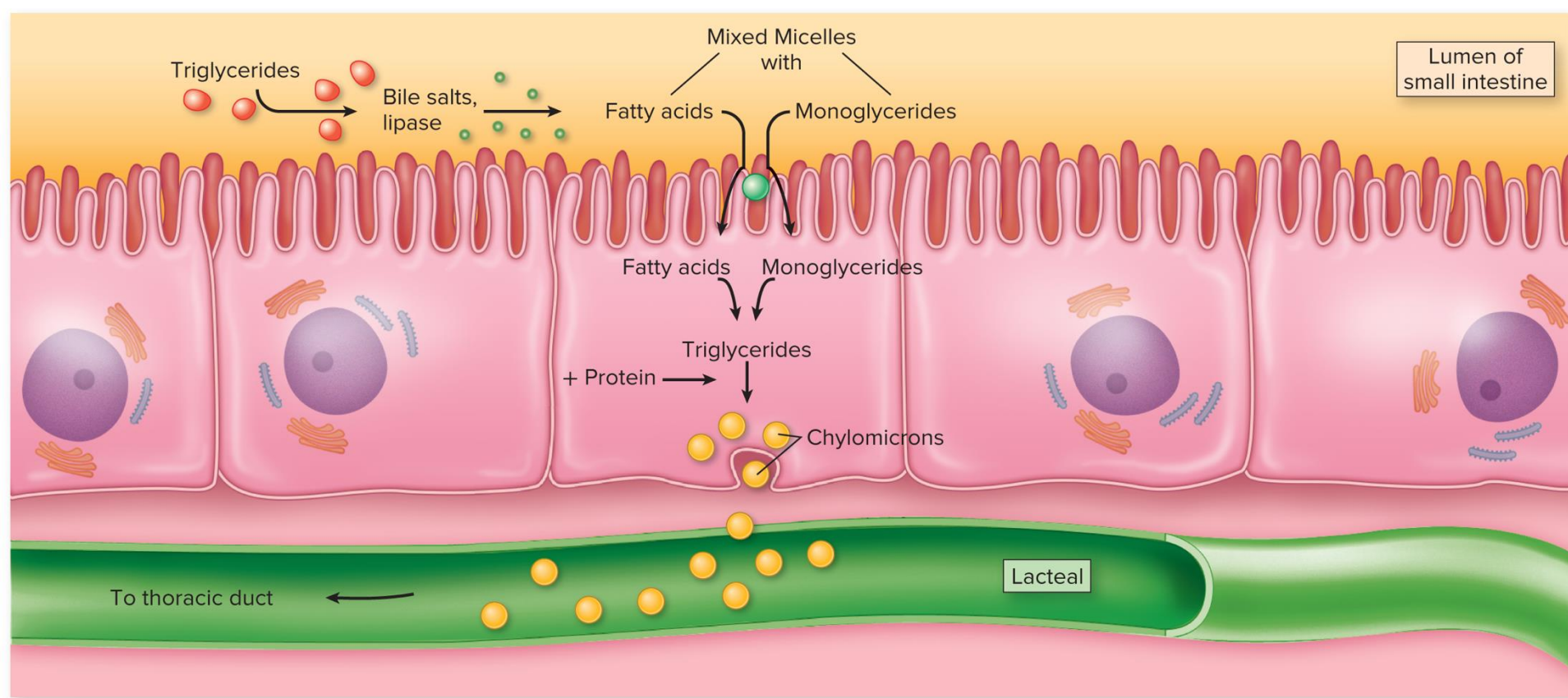
emulsification then digestion



- Liver and gall bladder release bile
- Bile salts emulsify large fat droplets
- **Pancreatic lipase** digests fat → monoglyceride + FFA's
- Monoglyceride + FFA's + bile salts → micelles

Lipid Emulsification, Digestion & Absorption

into lymphatic capillaries = lacteals



- FFA's / monoglycerides leave **micelles** → intestinal cell
- TG's reassembled + protein → **chylomicron**
- Chylomicron exocytosed → lacteals (*lymphatic capillary*) → blood²⁷

Question

What 2 processes are necessary for the breakdown of triglycerides so that they can be absorbed in the small intestine?

1. lipolysis and emulsification
2. lipolysis and glycolysis
3. emulsification and digestion
4. emulsification and anabolism

Digestive Enzymes

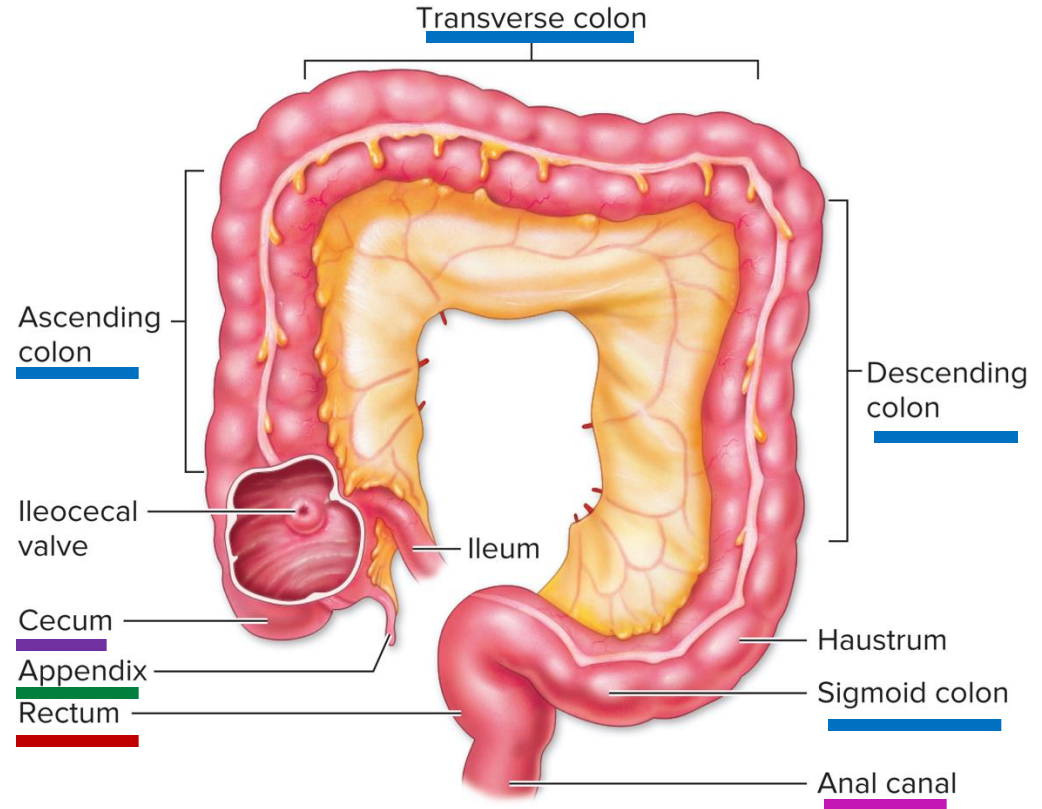
summary

Organ	Enzyme	Substrate
Salivary gland	amylase	starch
Stomach	pepsin	protein
Pancreas	amylase	oligosaccharide
	trypsin, chymotrypsin carboxypeptidase	polypeptide, peptide
	lipase	triglyceride
Small intestine <i>brush border</i>	lactase, sucrase, maltase	disaccharides
	aminopeptidase	peptide
Intestinal cell	di- & tri-peptidase	di- & tri-peptide ²⁹

Large Intestine Functions

appendix, cecum, colon, rectum, anal canal

- Absorption of salts & water → solidify feces
- Microbiota
 - *Provide vitamin K & B vitamins*
 - *Digest fiber*
 - *Aid in water reabsorption*
 - *Support immunity*
 - *Support normal weight*
 - *Etc., etc., etc.*



Accessory Organs

Salivary Glands: *secretes saliva*

- *Contains salivary amylase*
- *Moistens food*

Pancreas: *secretes many digestive enzymes*

Liver: *next...*

Oral cavity

Salivary glands

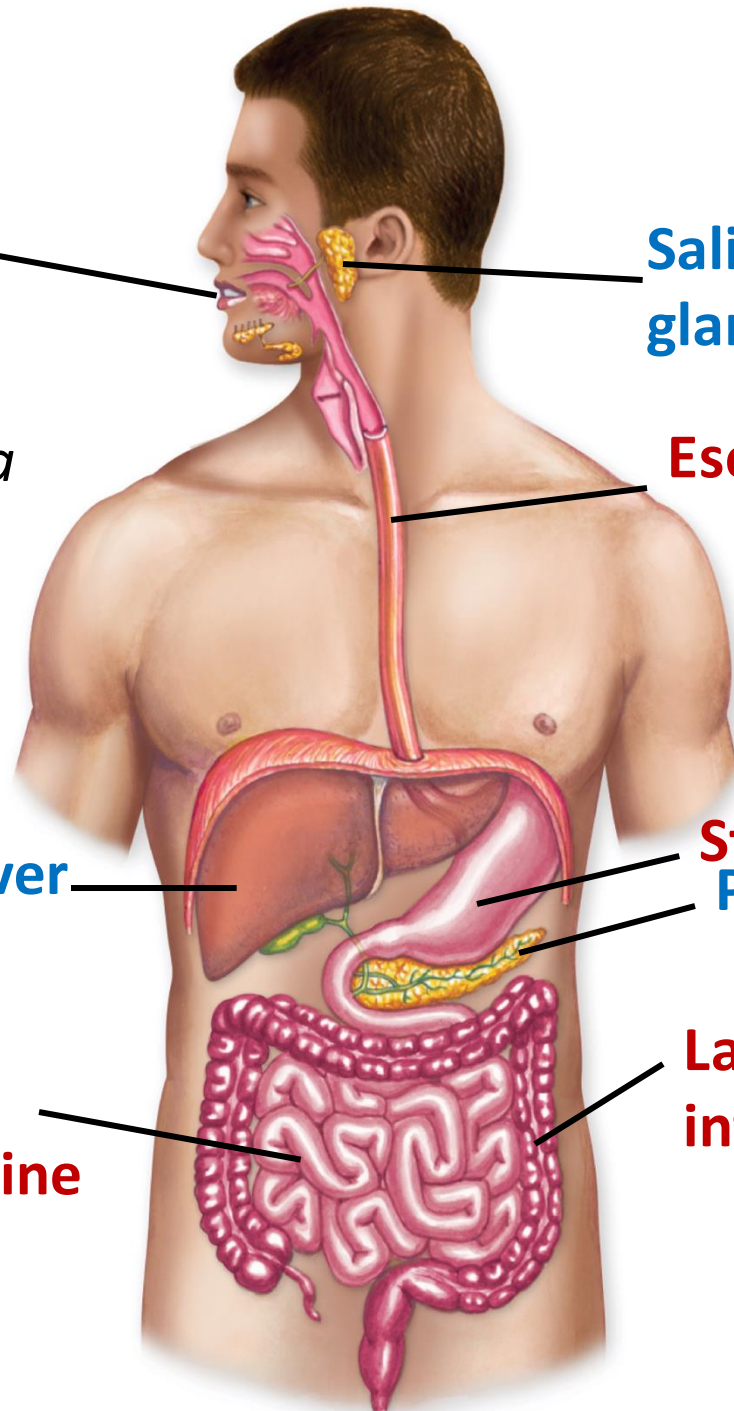
Esophagus

Liver

Stomach
Pancreas

Small intestine

Large intestine



Accessory Organ: Liver

Functions: bile production, biotransformation, phagocytosis

Bile production & secretion

Biotransformation:

- **Detoxification of blood**
chemically modify toxic chemicals to less toxic forms for excretion (ammonia/ NH_3 to urea, xenobiotics & steroid hormones to soluble forms → excreted by kidneys)
- **Transformation of metabolites to use**
glycogenolysis, gluconeogenesis, ketogenesis
- **Transform metabolites to store**
glycogenesis, lipogenesis

Accessory Organ: Liver

Functions: bile production, biotransformation, phagocytosis

Biotransformation (*cont.*):

- **Synthesis of plasma proteins**
fibrinogen (clotting); albumin (osmosis); angiotensinogen
- **Vitamin D synthesis**
skin → liver → kidneys
- **Recycling of hemoglobin**
save Fe → new hemoglobin; heme → bilirubin → bile

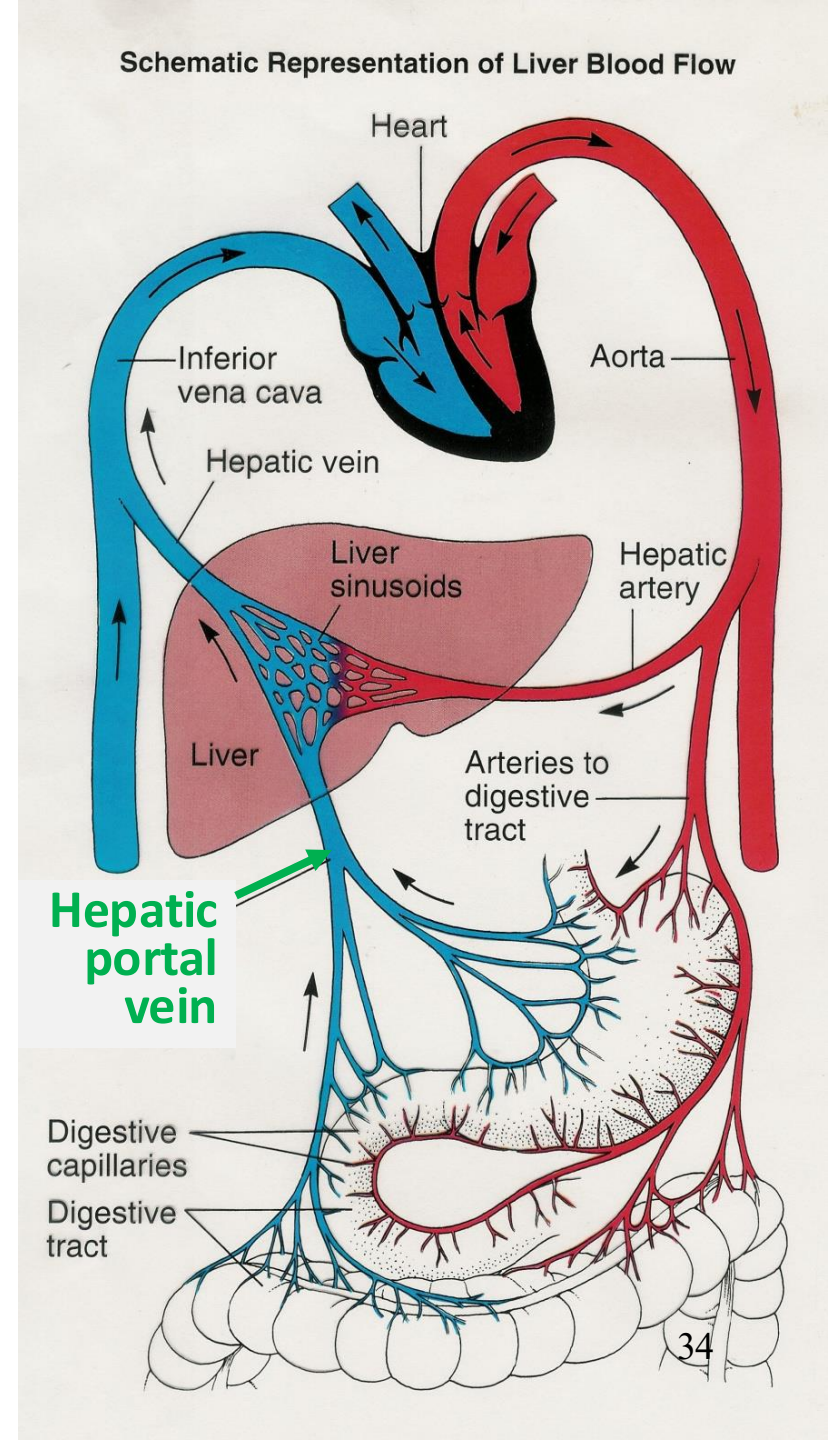
Phagocytosis

- Kupffer cells (*macrophages*): bacteria & old RBCs

Portal Circulation

2 capillary beds in a row

- 1. Endocrine system:** *connects hypothalamus & anterior pituitary capillary beds*
- 2. Urinary system:** *connects glomerulus & peritubular capillary beds*
- 3. Digestive system:** **Hepatic portal vein** connects stomach, intestines, pancreas & spleen capillaries to liver capillaries



Fluid Balance Sheet

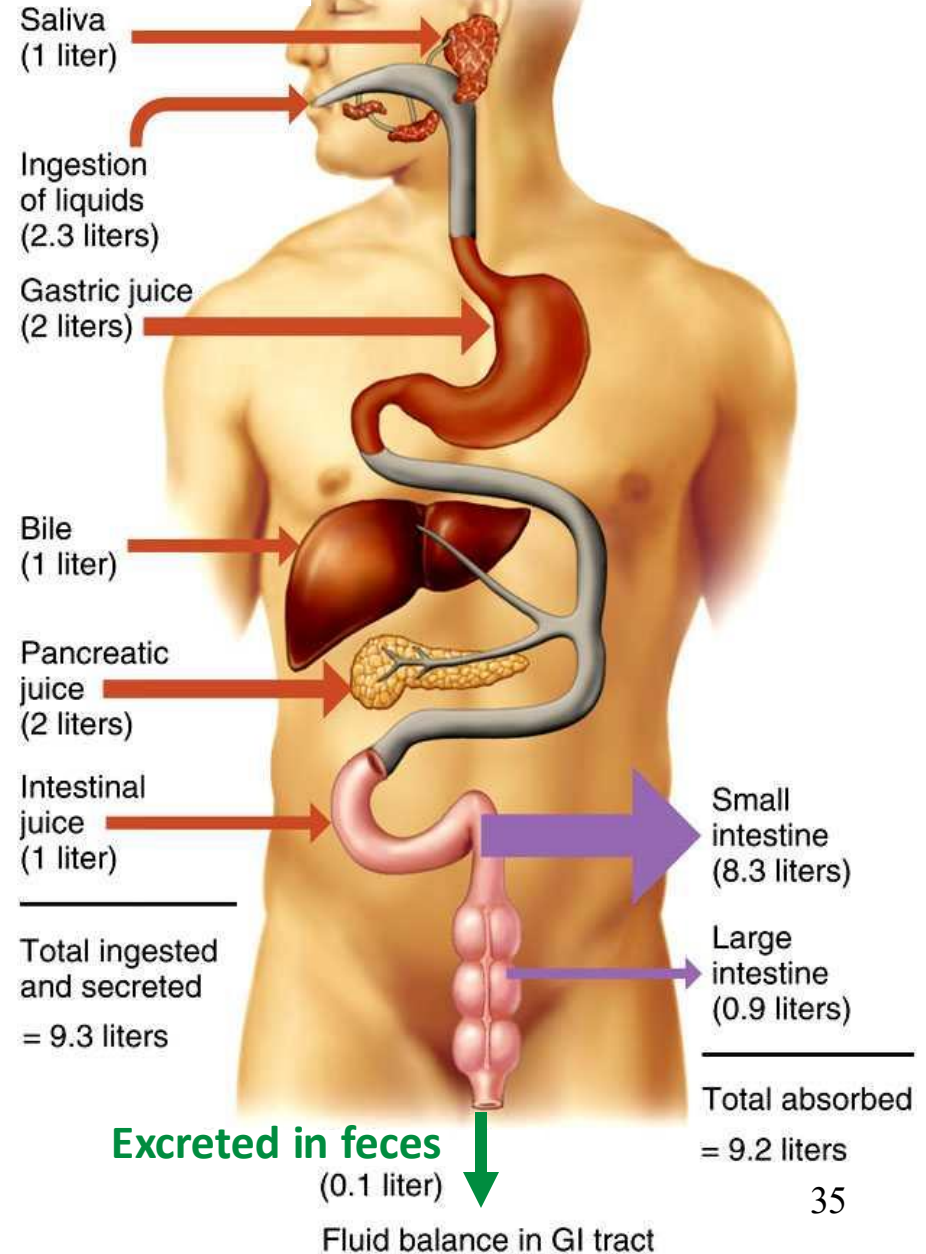
important clinically

Diarrhea

- Leading cause of child mortality & morbidity in the world.
- Results primarily from contaminated food & water.
- *From the World Health Organization (WHO)*

Ingested & Secreted

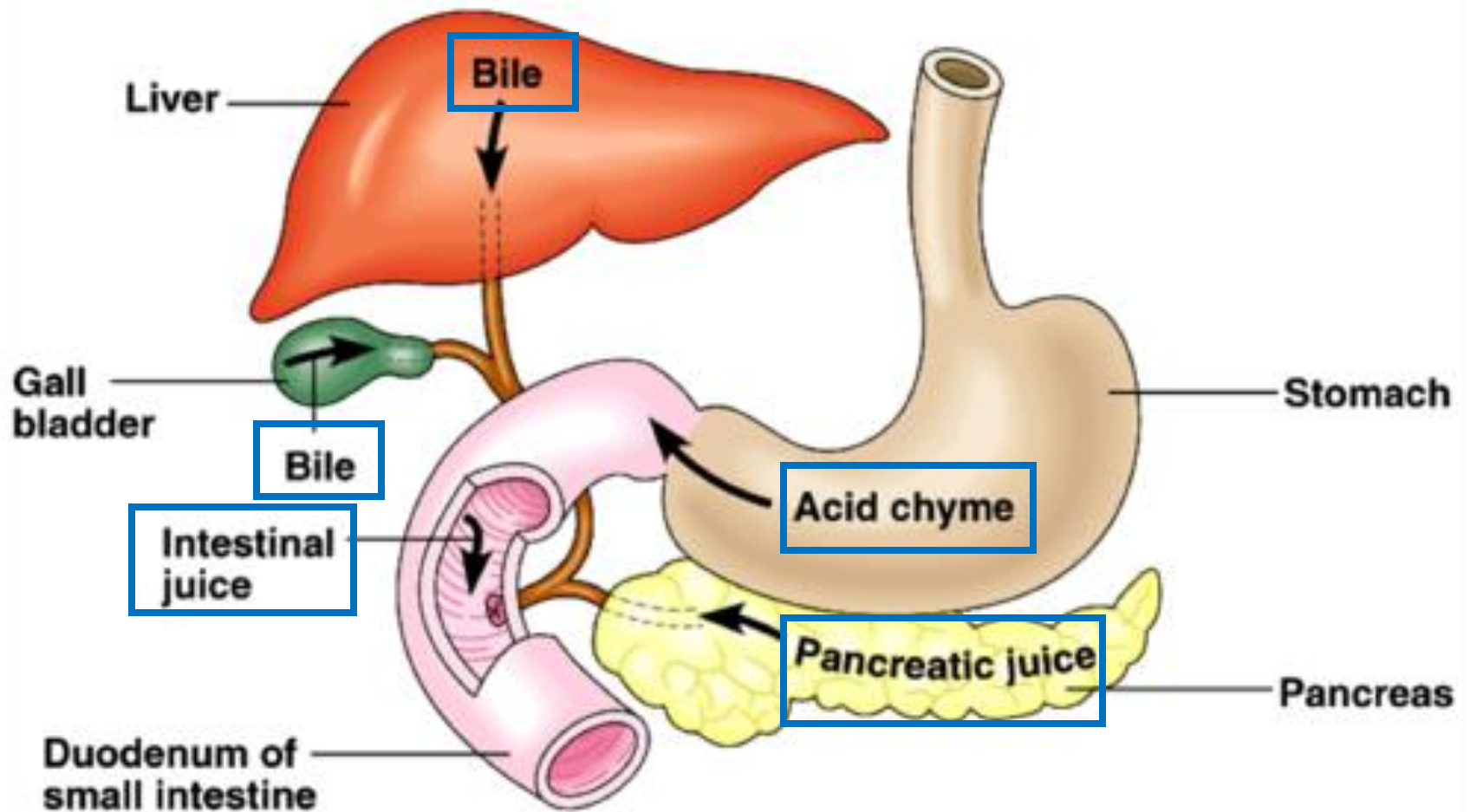
Absorbed



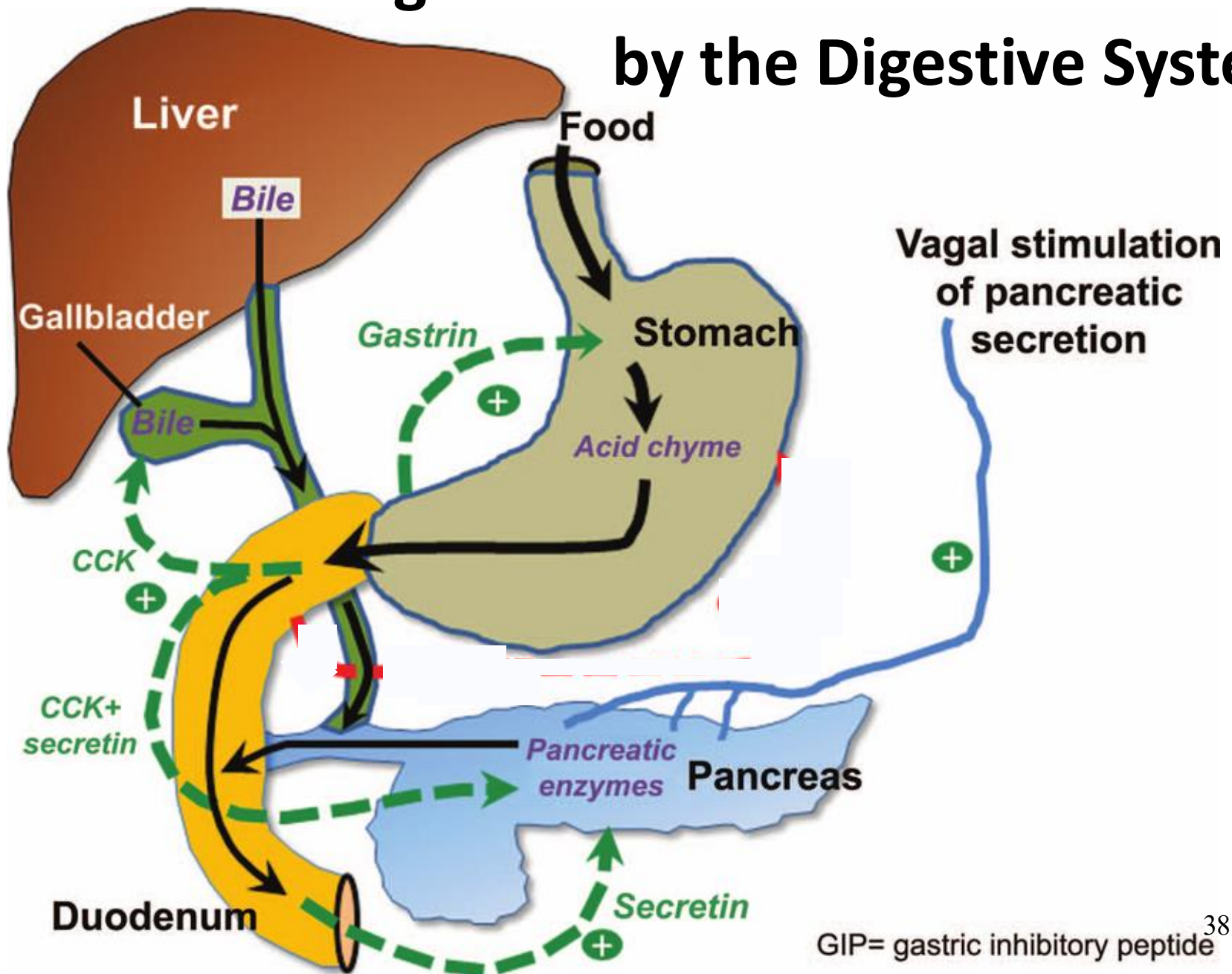
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Secretions into Duodenum



Regulation of **Hormones** Secreted by the Digestive System



Hormones (*paracrine*)

Gastrin:

- **Origin:** stomach
- **Stimulus:** *protein* in stomach
- **Response:** G cells (stomach) secrete gastrin → blood → stomach
 - Parietal cells → secrete *HCl*
 - Chief cells → secrete *pepsinogen*
 - *pepsinogen + HCl* → *pepsin*

Secretin:

- **Origin:** small intestine (duodenum)
- **Stimulus:** *HCl from stomach* in small intestine
- **Response:** tells pancreas (duct cells) → *secrete bicarbonate rich fluid* → duodenum (*neutralize stomach HCl*)

CCK: (cholecystokinin) (*chole "bile"; cysto, "sac"; kinin, "move"*)

- **Origin:** small intestine (duodenum)
- **Stimulus:** *fat* in small intestine (assume other nutrients there too)
- **Response:**
 - Tells pancreas (acinar cells) → *secrete all digestive enzymes*
 - Tells gall bladder → *secrete bile*